

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2024

Programme: BE

Full Marks : 100

Course: Engineering Economics (New)

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you understand about independent and dependent projects? 5
b) Define the inflation. What are its causes? 5
c) How do you understand the economic service life of a machine? 5
2. a) Define time value of money. Suppose that you make the monthly deposits of Rs 5,000 each into a bank account that pays an interest rate of 8% compounded weekly for 5 years. After 5 years, interest rate changes to 6% per year, how much money will you have accumulated in this bank account at the end of 8 years? 8
b) What is Supply & demand? Explain about principles of engineering economics. 7
3. a) Compute IRR and suggest the investment decision for the project given below: 8
Initial investment: Rs 90000
Annual Revenue: Rs 27000
Annual Expenses: Rs 8000
Salvage Value: Rs 10000
Expected Life: 6 year
MARR: 15%

OR

Find the simple and discounted payback periods and justify investment with the given cash flow information.

Initial investment: Rs. 400000

Annual benefit: Rs. 150000

Annual Cost: Rs. 30000

Salvage value: Rs. 100000

Useful life: 5 years

Interest rate: 10% per year.

- b) Explain about manufacturing cost and non manufacturing cost. 7
4. a) Using co-terminated assumption recommend the best project taking study period as 5 years. 8

Project	A	B
Initial Investment (Rs)	3,50,000	5,00,000
Annual Revenues (Rs)	1,30,000	1,75,000
Annual cost (Rs)	15,000	25,000
Salvage value (Rs)	35,000	50,000
Useful life	5 years	8 years
MARR	10%	

OR

Evaluate the following two feasible investment A and B having different useful lives, if MARR is 10% per year. Use PW method with repeatability assumption.

<u>Project: A</u>	<u>Project: B</u>
Initial investment: Rs 50000	Initial investment: Rs 150000
Annual Revenue: Rs 25000	Annual Revenue: Rs 70000
Annual Expenses: Rs 3000	Annual Expenses: Rs 2000
Salvage Value: Rs 15000	Salvage Value: Rs 40000
Expected Life: 3 year	Expected Life: 5 year

- b) Explain different project funding mechanism in Nepal for public projects. 7
5. a) Perform Sensitivity analysis using PW method over a range of (+ or -) 20% in initial investment, net annual revenue and salvage value. 8

Initial investment: Rs. 200000

Annual Revenue: Rs. 50000

Annual Expenses: Rs. 5000

Salvage Value: Rs. 25000

Expected Life: 10 year

MARR: 12% per year

- b) Suppose a company has decided to purchase a new machine having initial cost of Rs 18,000 by expecting 8 years useful life & have operating cost of Rs 3000 in 1st year. For remaining years, operating cost increase each year by 15% over previous year's operating cost & salvage value declines each year by 20% from the previous year's 7

salvage value. Find the economic service life of this new machine when MARR is 12% per year.

6. a) Define depreciation. What are the causes of it? If a machine costing Rs. 1,50,000 is purchased by expecting salvage value of Rs. 40,000 at the end of 6th year. Calculate depreciation amount for each year by
- i. SOYD
 - ii. Declining balance.
- b) Evaluate whether the project is feasible for investment or not using different types of BCR? 7
- Initial investment: Rs. 400000
Annual benefit: Rs. 75000
Annual Disbursement: Rs. 25000
Salvage value: Rs. 100000
Useful life: 10 years
Interest rate: 10% per year.
7. Write short notes on: (**Any two**) 2×5
- a) Depreciation by Sinking Fund method
 - b) Monte Carlo Simulation
 - c) Simple, Compound and Nominal Interest